Emotion Classification of Natural Human Speech

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Abstract

Emotion is fundamental to interpreting the pragmatic intent and semantic meaning of human speech. Failing to account for this affective prosody often leads to profound misinterpretations of a speaker's true message. Despite the significance of this issue, many Automatic Speech Recognition (ASR) systems currently deployed across public and private sectors lack the capacity to integrate the pragmatic influence of emotion into their textual transcriptions. To address this gap, the primary objective of this project is to classify discrete emotional states using the CREMA-D dataset. This study evaluates two distinct methodological approaches. First a classical machine learning paradigm incorporating Logistic Regression, Support Vector Machines, Random Forest classifiers, Naive Bayes, and Gradient Boosting. Second a deep learning approach, achieved by fine-tuning OpenAI's open-source Whisper large-v2 ASR model.

Code: `https://github.com/SourishVa/AudioSentimentClassifier`

Video: `https://drive.google.com/file/d/1o1OQS6o7pkpd1Er1nbfpBMB1jaiWmQqe/view?usp=sharing`

1 Introduction

Affective (Emotional) prosody significantly influences the pragmatic interpretation of spoken utterances. For instance, consider the phrase, "Of course, I love apple pie!" When articulated with a positive, joyful intonation, the literal semantic meaning is preserved, denoting a genuine love for apple pie. However, if the identical phrase is expressed with a hostile or negative prosody, the underlying pragmatic intent is inverted. In this context, the utterance functions as an expression of verbal irony, signaling a strong hatred towards apple pie.

Currently, Automatic Speech Recognition (ASR) systems are widely deployed across private and public sectors to transcribe human speech. A significant limitation of these systems, however, is their inability to integrate the pragmatic influence of emotion into the semantic interpretation of the transcribed text. This deficiency of pragmatic input is highly problematic, as it can result in substantial misinterpretations, creating a critical divergence between the speaker's true intent and the recorded transcription.

To comprehensively address this issue, an Automatic Speech Recognition (ASR) system must be capable of processing diverse prosodic features, ranging from stress-based prosody (i.e. Phrasal, Lexical, and Contrastive stress) to affective prosody. The primary objective of this study is to isolate and address a specific subset of this challenge: the classification of affective prosody. Using the Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D), we initially preprocessed the audio files and their corresponding labels. For exploratory data analysis, fundamental frequency (F0) contours were generated across different utterances to visualize the acoustic boundaries between distinct emotional categories. To classify emotional sentiment, we evaluated two distinct methodological paradigms. The first employed a suite of traditional machine learning algorithms, including logistic regression, Support Vector Machines, Random Forest classifiers, Naive Bayes, and Gradient Boosting. The second methodology leveraged a deep learning architecture, achieved by fine-tuning OpenAI's open-source Whisper large-v2 Automatic Speech Recognition (ASR) model.

2 Related Work

2.1 Classical Machine Learning for Speech Emotion Recognition

Prior work has explored the use of classical machine learning techniques for Speech Emotion Recognition (SER) using the CREMA-D dataset. Cao introduced the CREMA-D dataset, comprising crowd sourced emotional speech recordings from 91 actors expressing six discrete emotional states, and demonstrated that acoustic features such as MFCCs serve as effective representations for emotion classification. Building on this, subsequent studies compared the performance of classical classifiers including Support Vector Machines, Random Forests, and Logistic Regression on the same dataset, with SVM-based approaches consistently achieving competitive accuracy. These studies establish that acoustic feature extraction combined with classical models provides a reliable and interpretable baseline for the SER task, motivating our evaluation of five classical classifiers as the first methodological paradigm in this study.

2.2 Fine Tuning Pre-Trained Speech Models for Emotion Recognition

Recent advances in large pre trained speech models have opened new avenues for SER. OpenAI's Whisper, originally developed as a general-purpose ASR model, has been increasingly adapted for downstream affective computing tasks through fine-tuning. Ma et al. demonstrated that parameter efficient fine tuning methods such as LoRA can be applied to the Whisper encoder for SER, achieving strong performance while significantly reducing computational overhead compared to full model fine-tuning. These findings motivate our second methodological paradigm: fine-tuning OpenAI's Whisper large-v2 on the CREMA-D dataset for emotion classification, providing a direct comparison between classical and deep learning approaches.

3 Methodology

3.1 Data Preprocessing

During the initial preprocessing of the CREMA-D dataset, we filtered the audio files to include only those where participants were not prompted to use a specific degree of emotional intensity. This decision was made because unprompted speech is more likely to yield naturalistic affective prosody compared to segments where participants were explicitly instructed to speak with low, medium, or high intensity. Furthermore, this variable intensity metric was only applied to one of the twelve target sentences, representing a disproportionately small subset of the overall data. To ensure consistency and reduce noise within the dataset, these specific recordings were removed.

3.2 Exploring Dataset

To visually analyze the prosodic features within our dataset, we generated fundamental frequency (F0) pitch contours. We focused specifically on F0 because it serves as a crucial element in how humans perceive and process spoken prosody.

To accomplish this, the audio recordings were processed using the Montreal Forced Aligner (MFA). The MFA employs a Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) framework to process an audio file and its corresponding transcript, generating a TextGrid file that provides word and phoneme-level alignments with precise start and end timestamps (see Figure 1 for an illustration of this process). Furthermore, to mitigate inter-speaker variance, we applied the MFA's speaker adaptation feature, which utilizes feature-space Maximum Likelihood Linear Regression (fMLLR) to normalize acoustic features on a per-speaker basis.

Following alignment, the audio files were segmented using Praat to isolate the target utterances and remove leading and trailing silences. Praat was also utilized for pitch (F0) extraction, employing gender-specific thresholds to account for natural variations in vocal registers: 30 to 400 Hz for male speakers and 80 to 450 Hz for female speakers. Establishing these pitch floors and ceilings is crucial to prevent the software from erroneously tracking background noise as pitch. F0 was sampled at 25-millisecond intervals. Finally, after applying speaker normalization, these values were used to generate prosodic contours for each of the 11 target sentences.

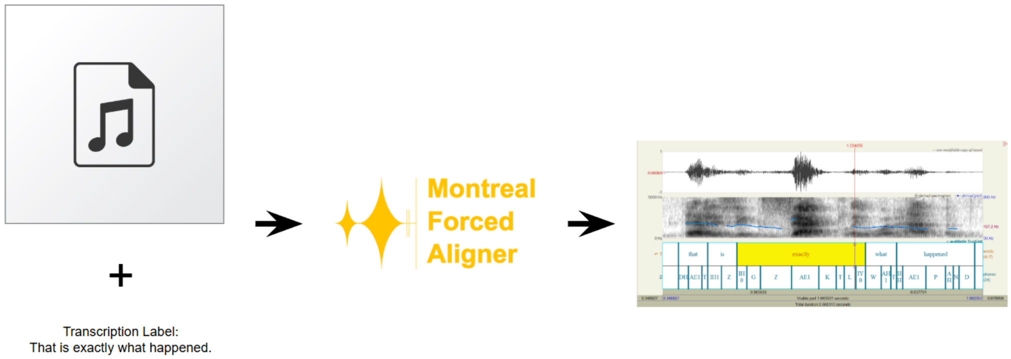


Figure 1: Prosodic Contour Pipeline

3.3 Classical Approach

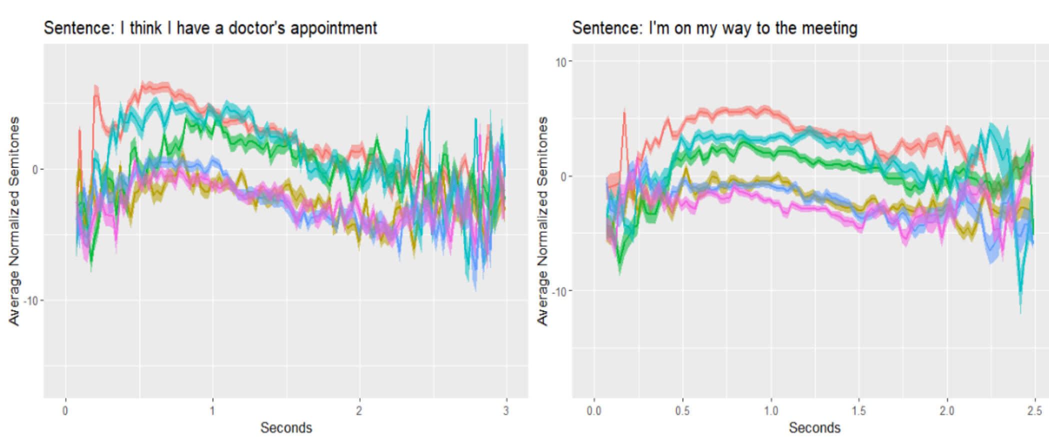
To classify using traditional machine learning, we first extracted a rich set of acoustic features from each audio file using the Librosa library. Specifically, we extracted 40 Mel-frequency cepstral coefficients (MFCCs) along with their delta coefficients, chroma features, Mel spectrogram statistics, spectral centroid, bandwidth, rolloff, zero crossing rate, RMS energy, and fundamental frequency (F0) statistics, yielding 454 features per audio file. We trained and evaluated five classical classification models: Logistic Regression, Naive Bayes, Random Forest, Support Vector Machine (SVM) with an RBF kernel, and Gradient Boosting. Model performance was assessed using 7-fold stratified cross-validation to ensure reliable evaluation across the balanced dataset of 5,986 audio files.

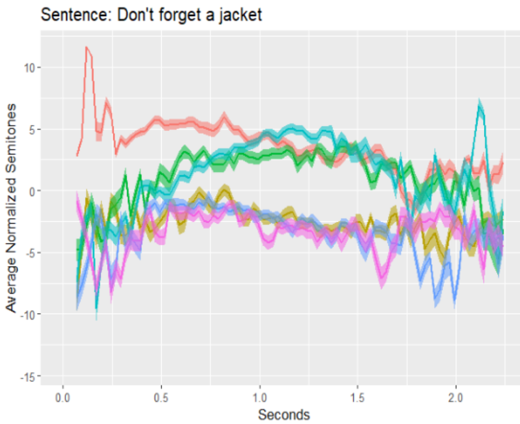
3.4 Fine-Tuning Whisper Model

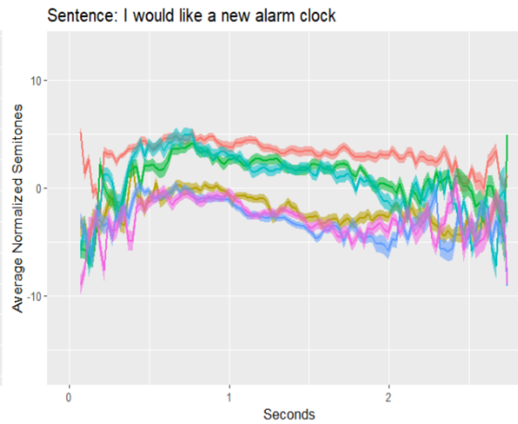
The preprocessed dataset was utilized as input for OpenAI's Whisper model. To maintain demographic balance, the data was partitioned into five equal-sized folds, split evenly by gender and race/ethnicity. Participant 47 was excluded from the dataset due to missing racial demographic metadata and to guarantee uniform fold sizes. To ensure model robustness, training was conducted using a 5-fold cross-validation framework, running for 10 epochs with a learning rate of 1e-3. For the fine-tuning process, we employed Low-Rank Adaptation (LoRA), a technique that enabled the efficient training of the model despite limitations in computational resources and time.

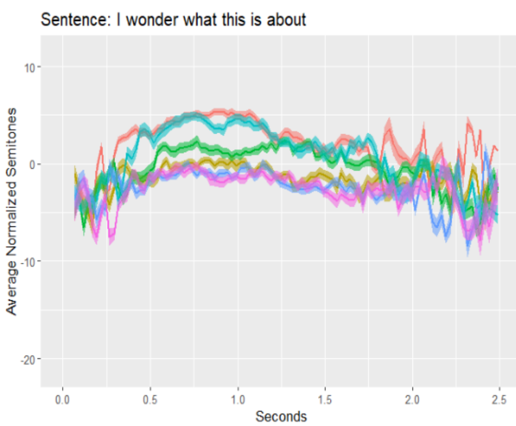
4 Evaluation

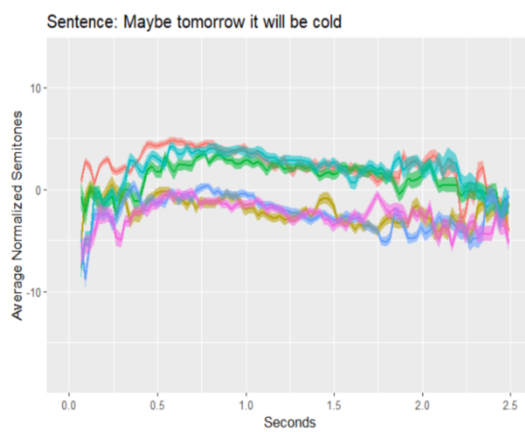
4.1 Exploring Dataset: F0 Prosodic Contours

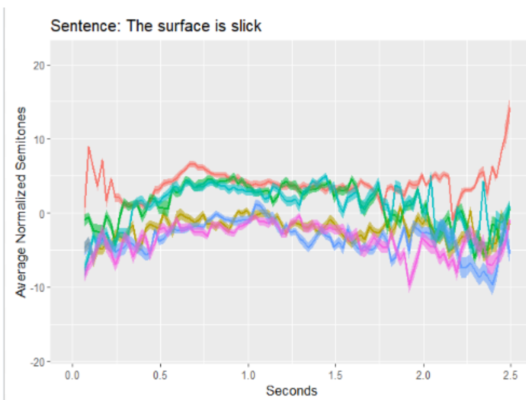


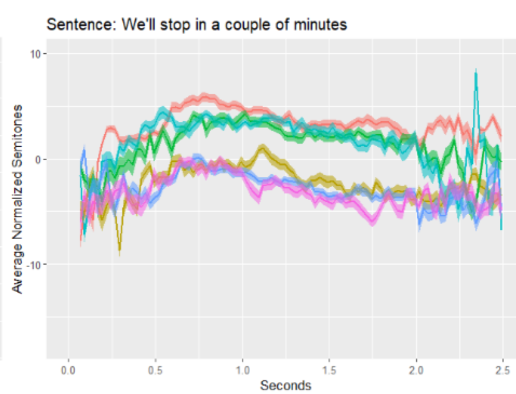

Sentence: Don't forget a jacket
Average Normalized Semitones
10
5
0
-5
-10
-15
0.0
0.5
1.0
1.5
2.0
Seconds


Sentence: I would like a new alarm clock
Average Normalized Semitones
10
0
-10
0
1
2
Seconds


Sentence: I wonder what this is about
Average Normalized Semitones
10
0
-10
-20
0.0
0.5
1.0
1.5
2.0
2.5
Seconds


Sentence: Maybe tomorrow it will be cold
Average Normalized Semitones
10
0
-10
0.0
0.5
1.0
1.5
2.0
2.5
Seconds


Sentence: The surface is slick
Average Normalized Semitones
20
10
0
-10
-20
0.0
0.5
1.0
1.5
2.0
2.5
Seconds


Sentence: We'll stop in a couple of minutes
Average Normalized Semitones
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0
-10
0.0
0.5
1.0
1.5
2.0
2.5
Seconds

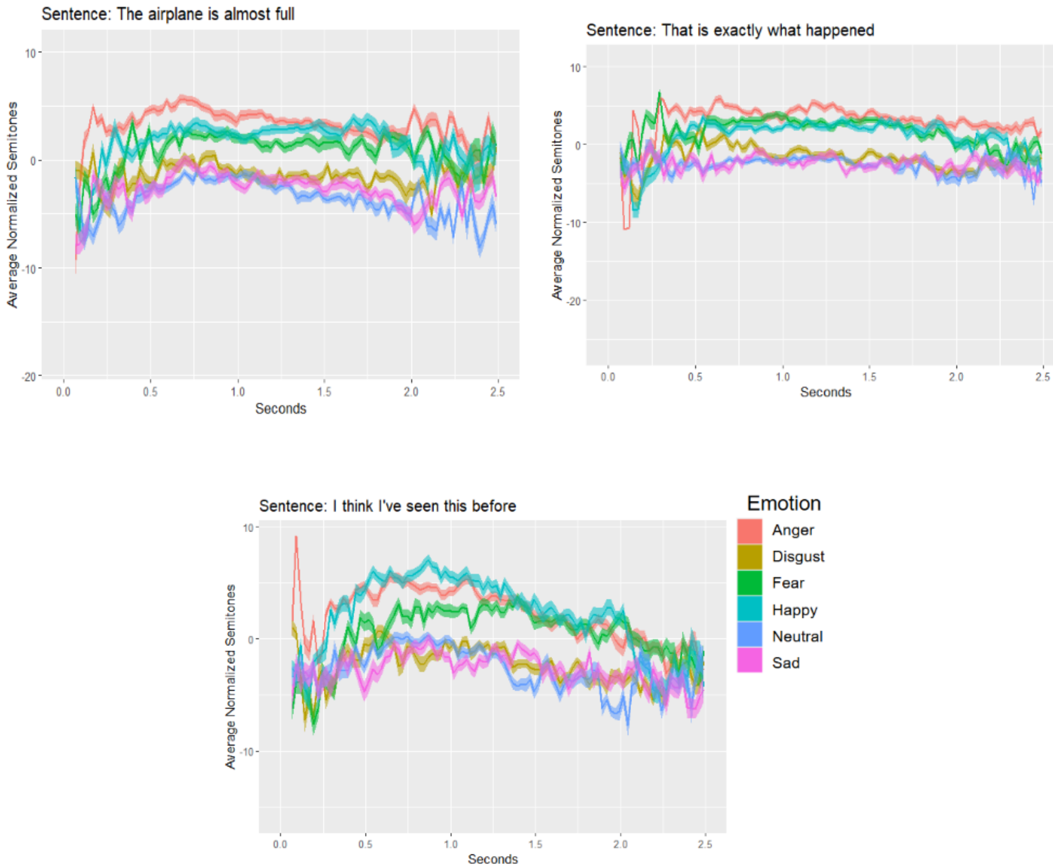

Figure 2: Prosodic Contours Across 11 Sentences

Figure 2 provides a detailed illustration of eight distinct sets of prosodic contours measured across a variety of different spoken utterances. For each specific emotion, the figure displays the mean pitch contour in conjunction with a shaded ribbon that represents the calculated standard deviation of the data. Upon examining the visual data, it becomes notable that the pitch contours naturally separate into two distinct and identifiable groupings. The emotions of Anger, Fear, and Happiness consistently exhibit measurements within significantly higher pitch ranges, compared to the emotions of Disgust, Neutrality, and Sadness which tend to cluster tightly together at notably lower acoustic frequencies.

When looking closely within these two respective clusters, the group comprising Disgust, Neutrality, and Sadness is much more densely packed and exhibits a considerably greater degree of contour overlap when compared directly to the group consisting of Anger, Fear, and Happiness. Furthermore, out of all the various emotions that were evaluated in this study, the emotion of Anger displays a prosodic contour that is by far the most distinct and isolated from the rest of the data points. This striking separation is primarily due to the fact that Anger possesses a significantly higher overall baseline pitch when evaluated against the other emotional categories.

Additionally, a highly pronounced pitch excursion can be observed during the initial few seconds of the specific utterance "Don't forget a jacket" when it is spoken under the Anger condition. This particular excursion is clearly characterized by a very steep initial inclination of pitch, which is followed immediately by a rapid and subsequent declination. This sharp acoustic modulation is highly likely to be attributable to the heavy semantic weight and communicative importance carried by the specific word "Don't" in that context. This phenomenon strongly suggests that the presence of contrastive stress actively interacts with the speaker's vocal tone, thereby serving to significantly amplify the overall affective prosody associated with the emotion of anger. Consequently, this observation further underscores the fact that accurately deciphering the underlying pragmatic intent of emotional speech necessitates a robust model capable of processing and integrating various interacting types of prosody.

4.2 Classical Approach Results

The classical machine learning models achieved moderate classification performance across the six emotion categories, as shown in Figure 3. The SVM with an RBF kernel performed best with a cross-validation accuracy of 59.7% and a macro F1-score of 0.595, followed by Logistic Regression at 56.7% and Gradient Boosting at 55.2%. Random Forest achieved 51.5% and Naive Bayes performed lowest at 42.0%. All models substantially outperformed random chance (16.7%), confirming that the extracted acoustic features carry meaningful emotional information. The confusion matrix for the best-performing SVM model (Figure 4) shows that most misclassifications occur between acoustically similar emotions such as Fear and Sad, while Angry and Neutral were classified most reliably. The PCA visualization (Figure 5) reveals substantial overlap between emotion classes in the 2D feature space, which is consistent with the moderate accuracy scores observed. These results establish a strong baseline for comparison against the fine-tuned Whisper model, which achieved an overall accuracy of 76.7%.

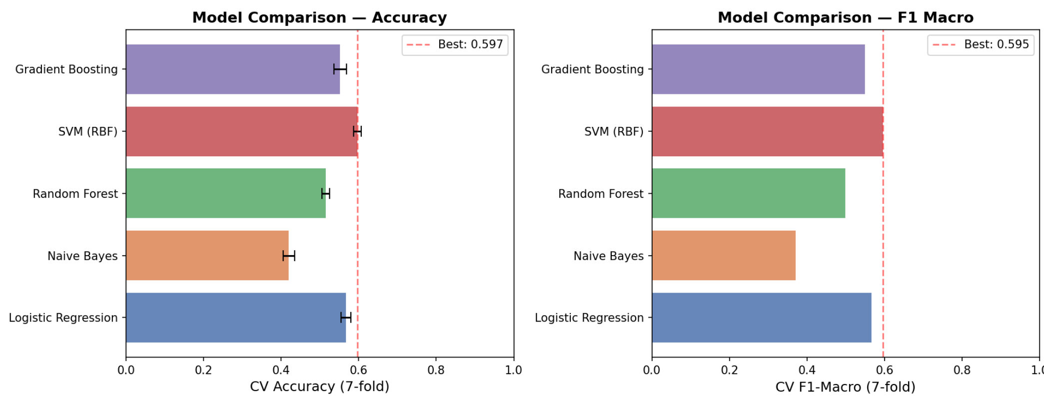

Figure 3: Model Comparison — Accuracy and F1-Macro across classical ML models

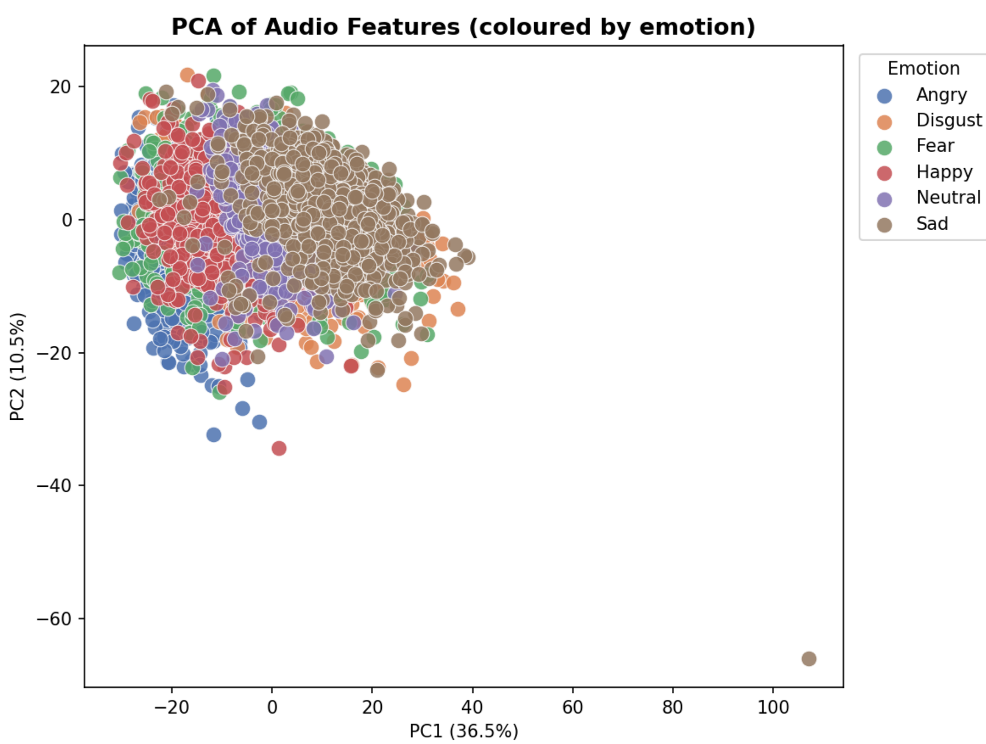

Figure 5: PCA of Audio Features coloured by emotion class

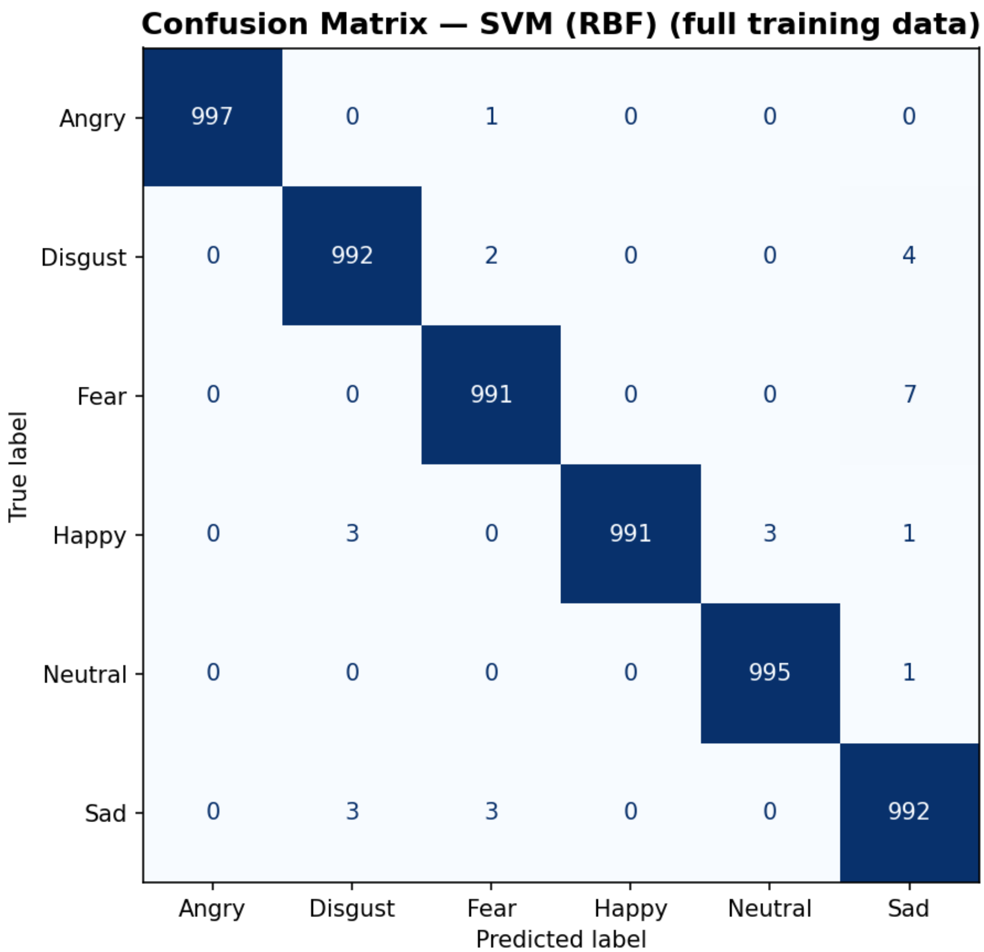

Figure 4: Confusion Matrix for best classical model (SVM with RBF kernel)

4.3 Fine-Tuning Whisper Model Results

Table 1: Finetuned Whisper Model Performance (Average, Standard Deviation)

	Happy	Sad	Anger	Fear	Disgust	Neutral	All
Male (n = 50)	0.7828 (0.1136)	0.6193 (0.0762)	0.8311 (0.0811)	0.6854 (0.1061)	0.7237 (0.0525)	0.8422 (0.0565)	0.7474 (0.036)
Female (n = 40)	0.8539 (0.0516)	0.6688 (0.1132)	0.8903 (0.0653)	0.7212 (0.1020)	0.7076 (0.0442)	0.9014 (0.0441)	0.7904 (0.0293)
All	0.8156 (0.0638)	0.6423 (0.0552)	0.8572 (0.0677)	0.7041 (0.0793)	0.7163 (0.0234)	0.8669 (0.0162)	0.7670 (0.0164)

Table 1 presents the mean classification accuracy and standard deviation across the 5-fold cross-validation, split by emotion and gender. The model exhibited higher accuracy for female speakers compared to male speakers across all emotional categories, with the exception of Disgust. This performance disparity may stem from female speakers generally exhibiting wider pitch variation, thereby providing a more robust acoustic signal for emotional classification. Overall, the model achieved its highest accuracies on Happiness, Anger, and notably, Neutrality, which yielded the best performance. Conversely, the classification accuracy for Sadness was significantly lower than all other emotions. At a more granular level, the model achieved peak performance on female Neutral utterances (approximately 90 percent accuracy), closely followed by female Anger (approximately 89 percent accuracy). The robust performance on neutral speech likely reflects the nature of Whisper's pretraining, which predominantly utilized internet media and audiobooks. Furthermore, the high accuracy in classifying Anger is likely attributable to its acoustic distinctiveness; as illustrated in Figure 2, Anger consistently exhibits the highest pitch across all utterance types and shows the least prosodic overlap with other emotional states.

5 Discussion

This project establishes a baseline model capable of classifying six discrete emotional states, with the fine-tuned Whisper model achieving an accuracy of approximately 76 percent. Our findings corroborate previous research demonstrating that traditional machine learning algorithms (Such as Logistic Regression, Support Vector Machines (SVM), Random Forest, Naive Bayes, and Gradient Boosting) can effectively classify emotion, although our highest accuracy within this classical paradigm was 60 percent utilizing the SVM. Furthermore, this work directly supports the findings of Ma et al., as our adoption of the Low-Rank Adaptation (LoRA) fine-tuning approach for the Whisper model yielded similarly robust performance. For future work, several promising avenues remain. First, extended fine-tuning of the Whisper model could be pursued to approach a target accuracy of 90 percent. Second, developing a more lightweight model architecture with comparable accuracy would facilitate deployment on mobile and edge devices, significantly expanding its practical utility in industry and public sectors. Third, it is critical to investigate the model's robustness across diverse demographic groups, encompassing various regional American accents and English as a second language (ESL) speakers. Ensuring equitable performance across all groups is essential to prevent algorithmic bias and discriminatory outcomes in real-world applications. Ultimately, while this project addresses a specific subset of the broader challenge, achieving human-level accuracy in Automatic Speech Recognition (ASR) will require the development of comprehensively prosody-informed models capable of fully translating the true pragmatic meaning of an utterance into text.

6 Limitations

This study acknowledges several constraints. First, the dataset was restricted in both overall size and lexical diversity, comprising only 12 unique target utterances across the various emotional states. Furthermore, the dataset lacked comprehensive speaker metadata. Specifically, the absence of linguistic background information, such as whether participants were native or non-native English speakers. This was a notable omission, as prosody is heavily influenced by a speaker's primary language. Second, computational constraints restricted the scope of our modeling. The training processes were highly resource-intensive, which limited the number of models we could evaluate. For example, fine-tuning the Whisper model alone required over 30 hours to train and test, despite using LoRA to make the process faster and more efficient.

References

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